MRI Image Quality Metrics (IQMs)

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December 9, 2025

Abstract

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Contents

Introduction

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1 Per-Slice IQMs (Foreground / Background pixels)

All metrics below accept foreground pixels $\mathbf{F}$ (from segmentation) and background pixels $\mathbf{B}$. NaN/Inf inputs yield NaN. For BG-dependent metrics (SNR1, SNR2, SNR4, CNR, CJV, FBER), if background is missing or unusable (empty, zero variance, non-finite), we fall back to foreground homogeneity μ_F/σ_F to avoid artificial NaNs.

1.1 Mean (MEAN)

Foreground mean intensity:

$$\mathrm{MEAN} = \mu_F = \frac{1}{N_F}\sum_i F_i$$

1.2 Range (RNG)

Foreground intensity range:

$$\mathrm{RNG} = \max(F) - \min(F)$$

1.3 Variance (VAR)

Foreground variance:

$$\mathrm{VAR} = \sigma_F^2 = \frac{1}{N_F}\sum_i (F_i - \mu_F)^2$$

1.4 Coefficient of Variation (CV)

Foreground coefficient of variation (percent):

$$\mathrm{CV} = \frac{\sigma_F}{\mu_F} \times 100$$

1.5 Contrast per Pixel (CPP)

Laplacian-filtered contrast, mean response of a 3×3 Laplacian:

$$f_1 = \frac{1}{8}\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}, \quad \mathrm{CPP} = \mathrm{mean}\left(f_1 * F\right)$$

where $*$ is 2D convolution (foreground is median-filtered only when used for PSNR below).

1.6 Peak Signal-to-Noise Ratio (PSNR)

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\max(F)^2}{\mathrm{MSE}(F,\hat{F})}\right)$$

where $\hat{F}$ is either the background $\mathbf{B}$ (if provided) or a median-filtered version of $\mathbf{F}$.

1.7 SNR1

Foreground/background std; fallback to foreground stability if background unusable [1], [2]:

$$\mathrm{SNR1} = \begin{cases} \dfrac{\sigma_F}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise (fg homogeneity)} \end{cases}$$

1.8 SNR2

Foreground mean over background std; fallback to foreground stability if background unusable:

$$\mathrm{SNR2} = \begin{cases} \dfrac{\mu_F}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.9 SNR3

Foreground standard deviation divided by centered-foreground standard deviation (stability check):

$$\mathrm{SNR3} = \frac{\sigma_F}{\sigma_{F-\mu_F}}$$

1.10 SNR4

Foreground mean over background mean; fallback to foreground stability if background unusable:

$$\mathrm{SNR4} = \begin{cases} \dfrac{\mu_F}{\mu_B}, & \mu_B \text{ finite and } \mu_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.11 Contrast-to-Noise Ratio (CNR)

Foreground/background mean difference over background std; fallback to foreground stability if background unusable:

$$\mathrm{CNR} = \begin{cases} \dfrac{\mu_F - \mu_B}{\sigma_B}, & \sigma_B \text{ finite and } \sigma_B \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.12 Patch Coefficient of Variation (CVP)

Foreground patch coefficient of variation:

$$\mathrm{CVP} = \frac{\sigma_F}{\mu_F}$$

1.13 Coefficient of Joint Variation (CJV)

Joint variation between foreground and background; fallback to foreground stability if background unusable:

$$\mathrm{CJV} = \begin{cases} \dfrac{\sigma_F + \sigma_B}{|\mu_F - \mu_B|}, & \text{finite and } |\mu_F - \mu_B| \neq 0 \\ \dfrac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.14 Entropy Focus Criterion (EFC)

Normalized Shannon entropy of foreground:

$$p_i = \frac{F_i}{\sqrt{\sum_j F_j^2}}, \quad \mathrm{EFC} = -\frac{1}{\log N_F} \sum_i p_i \log p_i$$

higher values suggest more ghosting/noise.

1.15 Foreground-Background Energy Ratio (FBER)

Median energy ratio; fallback to foreground stability if background unusable:

$$\mathrm{FBER} = \begin{cases} \frac{\mathrm{median}(F^2)}{\mathrm{median}(B^2)}, & \text{finite and median}(B^2) \neq 0 \\ \frac{\mu_F}{\sigma_F}, & \text{otherwise} \end{cases}$$

1.16 Skewness (SKW)

Foreground skewness (third standardized moment):

$$\mathrm{SKW} = \frac{\frac{1}{N_F} \sum_i (F_i - \mu_F)^3}{\sigma_F^3}$$

1.17 Excess Kurtosis (KURT)

Foreground excess kurtosis (fourth standardized moment minus 3):

$$\mathrm{KURT} = \frac{\frac{1}{N_F} \sum_i (F_i - \mu_F)^4}{\sigma_F^4} - 3$$

2 Participant-Level QC

2.1 Failure Count (FAIL_FRAC)

Counts total non-finite IQM evaluations (NaN/Inf) across all slices/metrics for a participant. Higher counts flag poor FG/BG segmentation or corrupted slices:

$$\mathrm{FAIL\_FRAC} = \sum_{s \in \text{slices}} \sum_{m \in \text{IQMs}} \mathbf{1}\{\mathrm{IQM}_{m,s} \text{ is non-finite}\}$$

References

[1] Amir Reza Sadri et al. "MRQy—An open-source tool for quality control of MR imaging data". In: *Medical physics* 47.12 (2020), pp. 6029–6038.

[2] Rafsanjany Kushol et al. "DSMRI: domain shift analyzer for multi-center MRI datasets". In: *Diagnostics* 13.18 (2023), p. 2947.